My experience as an intern of the Rosa Sensat internship

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1 Abstract

Hello, my name is Jan. A few months ago, I was granted the opportunity to participate in the Rosa Sensat Internship at the Computer Vision Center (CVC). The aim of the internship was to familiarize me with the research and development of new technologies. Although the internship was heavily focused on AI, specifically computer vision, the knowledge I have gained during these three months will undoubtedly be applicable to many other fields.

During my stay at the CVC, I've taken part in a cloud segmentation project. The objective was to analyze Sentinel 1 or 2 multispectral images and accurately identify different types of clouds, including thick clouds, thin clouds, no clouds, or cloud shadows. Subsequently, I compared my results with those obtained from previously trained models.

2 Experience

Before we start, there is one thing you have to understand about me, that is that I started the internship from zero, meaning that I hadn't done any IA projects before.

This was a big concern for me at the beginning of the internship. However, as the internship progressed, these worries gradually dissipated. This was partly due to my dedication and investing a couple of learning hours, but mainly it was because of the comprehensive nature of the CVC and especially my mentor, as they tried to teach me at a pace that suited my level of understanding the most.

As I took off with the project, my initial task was to get familiar with the dataset (DS), specifically the CloudSen12 DS. This DS consists of a meticulously classified set of Sentinel 1 and 2 multispectral images for cloud segmentation. It also offered lots of well-organized and categorized images, along with numerous examples of use cases. Additionally, the paper explains fully the development of the CloudSen12 team's work.

This was maybe too big for me, and, as it was my first time, I didn't know what I was looking at or what my goal was. As a result of this, I did some Google colabs, some running examples, some of my code, and some were a combination of both. This didn't make the project advance that much, but it did help me understand various things that would later prove useful.

Once I understood the principles of the DS, I moved on to the model. My first model was a simple U-Net, which, using different convolution layers, was capable of downsampling images and then upsampling them while extracting different patterns from them.

Selecting an appropriate loss function posed a considerable challenge for me, given the enormous amount of options available. Initially, I opted for binary cross-entropy. However, as my mentor wisely pointed out, that would only be useful in case I tried to extract two classes, whereas my objective was to segment four of them.

Ultimately, I decided to settle for the "categorical-crossentropy-loss" function. However, at the time of writing this report, I am questioning its effectiveness and exploring alternative options, such as the Dice Loss or the IoU.

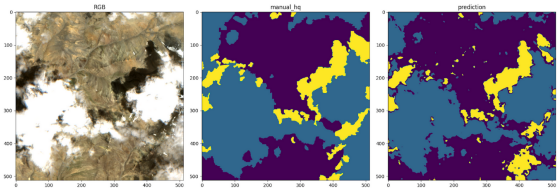


Figure 1: One of my first succesful atepmpts

After having trained the model for the first time, I experienced mixed emotions regarding the results. On one hand, I felt a sense of satisfaction for achieving such a significant portion of the project and because some of the resulting masks appeared to be reasonably satisfactory. However, on the other hand, I was a bit skeptical of the results.

This is due to the fact that, while some masks seemed to align with the image they came from, they did not align that well with the ground truth.

After conducting a few not-so-thorough investigations, I identified the problem: I had been utilizing only the RGB bands, whereas the images actually came in a wide variety of bands. Depending on the satellite, there could be up to 13 bands available. This was only one of the many rookie mistakes I made along the way. It originated from a lack of reading, understanding, and experience.

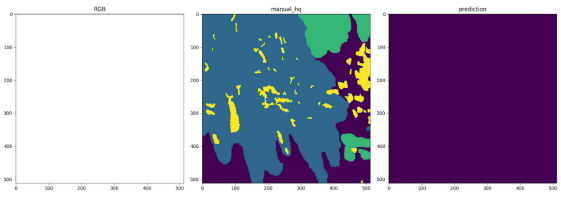


Figure 2: A suspicious image

Upon resolving one problem, others inevitably emerge. For instance, when identifying cloud shadows, certain images with steep mountains projecting shadows pose a challenge for our model. Similarly, fields of corn or wheat can sometimes be mistaken for clouds, leading to confusion and the appearance of small, square-like clouds on the resulting mask, where not even a cloud is supposed to be present. These are all problems we're currently trying to resolve.

Aside from all the experimentation with various models, loss functions, and analyzing results, I've also developed a GitHub repository containing detailed functions, scripts, and code snippets aimed at assisting future individuals working on similar projects. Within the repository, I also provide a brief overview of the dataset, the model, and the project itself.

3 Conclusions

Looking back now, I realize how many silly or rookie mistakes I've made: not using all the bands, not getting more images, not making the model bigger, not realizing the memmaps were already made... All those tiny things that would not have slipped through an experienced mind but did to mine have cost me great deals of time.

Having said that, I don't think that what I did was bad, in the sense that it's my first time, and I was doomed to make mistakes. Fortunately, I did plenty of them, and I will leave this internship with a big bag of knowledge on my back.

From a time point of view, although at times it's been hard to combine this internship with my studies, I would repeat this experience anytime, and I recommend anyone do so as well. For those more curious, I'm currently making a double degree in software engineering and telecommunications engineering, which is very time-demanding, and if you are studying something different and less time-consuming, I encourage you even more to look into this opportunity and similar ones, for this is a very fruitful experience.

Lookning forward I think this has just been an introduction, a preface, and that there is still a lot for me to learn. Lucky for me, I am eager to continue, to get new experiences, to participate in new projects, to meet new and wiser people, to share my knowledge, to make new mistakes, and, ultimately, to learn and let learn.